科学营　第一讲课后练习

1. (1) $(t+2)(t^2+t+1)$　(2) $(x-2y-1)(x-2y-2)$

(3) $(\frac{1}{3}y+x^n-\frac{1}{2})(\frac{1}{3}y-x^n+\frac{1}{2})$　(4) $-3a^n(a-3b^2)^2$

(5) $(a+b+c)(a+2b+3c)$　(6) $(x-2y+z)^2$

(7) $(x^2+4x+6)(x^2+8x+6)$　(8) $(x-1)^2(y-1)^2$

(9) $(x-4)(2x+7)(2x^2-x-8)$　(10) $(x^2-xy+y^2)^2$

(11) $(x+1)(y+1)(xy+x+y-1)$　(12) $(x-1)(x+1)(x-2)(x+4)$

(13) $-(x-y)(y-z)(z-x)(x+y+z)$　(14) $5(x+y)(y+z)(z+x)(x^2+y^2+z^2+xy+yz+zx)$

(15) $-(x-y)(y-z)(z-x)(x+y)(y+z)(z+x)$　(16) $4abc^2$

(17) $(x+y)(x-z)(y-z)$　(18) $2(x+2)(x+6)(x^2+8x+26)$

(19) $(x+1)(2x-1)(x^2+4)$　(20) $(5x+7)(x^3+x^2+2x-1)$

(21) $24xyz$　(22) $(x+2y)(x-3y)(x^2-2xy+4y^2)(x^2+3xy+9y^2)$

(23) $(x+a+b)(x+a-b)(x-a+b)(x-a-b)$　(24) $(x+y)^2(x-y)^2(a+b)^2(a-b)^2$

(25) $(x^2+xy+y^2)(a^2+ab+b^2)$　(26) $(x+1)(x-1)(xy-x+y+1)(xy-x-y-1)$

(27) $3abxy(a+b)(x+y)$　(28) $(x+2)(x+6)(x^2+8x+10)$

(29) $a(a-5)(a^2-5a+10)$　(30) $(x-1)(x+2)(x-3)(x+4)$

(31) $(x+2)(x+4)(x^2+5x+8)$　(32) $(x-1)(x+2)(x^2+x+5)$

2. O　原式 $=(x+y)(x^2+y^2)$

3. 左右两边同乘2，配方得 $(a-b)^2+(b-c)^2+(c-a)^2=0$

4. 证法1：$(a^2+b^2)^2=a^4+b^4+2a^2b^2=9a^2b^2$　证法2：$(\frac{a}{b})^2+1=3(\frac{a}{b})$，$b\neq0$时，可解得$\frac{a}{b}$

5. 证法1：$n(n^2-1)$被3！整除，分析$(n^2-5n+26)$模5的余数　证法2：原式$=(n-3)(n-2)(n-1)n(n+1)+20(n-1)n(n+1)$

6. 证法1：原式$=(x^2+ax+b)(x^2+cx+d)$，对应系数相等列方程组　证法2：原式$=(x^2+ax+1)(x^2+bx-1)$

7. 题干有误，应为 $B=x^2y+xy^2+y^2z+yz^2+z^2x+zx^2+2xyz$　证法：$x+y=0$时$A=B=0$，推出$B=(x+y)(y+z)(z+x)$是A因式